Toward Stable Semi-Supervised Remote Sensing Segmentation via Co-Guidance and Co-Fusion

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***Abstract*—Semi-supervised remote sensing (RS) image semantic segmentation offers a promising solution to alleviate the burden of exhaustive annotation, yet it fundamentally struggles with pseudo-label drift, a phenomenon where confirmation bias leads to the accumulation of errors during training. In this work, we propose Co2S, a stable semi-supervised RS segmentation framework that synergistically fuses priors from vision-language models (VLMs) and self-supervised models (SSMs). Specifically, we construct a heterogeneous dual-student architecture comprising two distinct ViT-based vision foundation models initialized with the weights of pretrained CLIP and DINOv3 to mitigate error accumulation and pseudo-label drift. To effectively incorporate these distinct priors, a semantic co-guidance mechanism is introduced that utilizes text embeddings and learnable queries to provide explicit and implicit class-level guidance, respectively, thereby jointly enhancing semantic consistency. Furthermore, a global–local feature collaborative fusion strategy is developed to effectively fuse the global contextual information captured by CLIP with the local details produced by DINOv3, enabling the model to generate highly precise segmentation results. Extensive experiments on six popular datasets demonstrate the superiority of the proposed method, which consistently achieves leading performance across various partition protocols and diverse scenarios. Project page is available at https://xavierjiezou.github.io/Co2S/**

***Index Terms*—Remote sensing (RS) images, semantic segmentation, semi-supervised learning, vision foundation models.**

I. Introduction

REMOTE sensing (RS) imagery has become indispensable for a wide range of Earth observation applications, including land-cover mapping, urban planning, ecological monitoring, and disaster assessment [1]. With the rapid advancement and widespread deployment of high-resolution satellite and aerial sensors, RS data now exhibit large spatial coverage [2], complex scene layouts, and fine-grained category structures. Consequently, pixel-wise semantic segmentation has emerged as a core task, requiring models to be simultaneously sensitive to high-level semantics (e.g., category identity and contextual relationships) and low-level details (e.g., object boundaries and small structures) [3], [4]. Nevertheless, the fully supervised deep models that currently dominate this field heavily depend on exhaustive pixel-level annotations, which are notoriously expensive and time-consuming to obtain due to the need for expert interpretation [5]. In practice, only a small fraction of available images can be labeled, while the vast majority remain unlabeled [6]. This scarcity has spurred a shift toward label-efficient paradigms across various RS tasks, such as unsupervised hyperspectral image restoration [7], [8]. Under such conditions, there is a pressing need for semi-supervised segmentation methods that can effectively exploit abundant unlabeled data, cope with label scarcity, and maintain stable optimization despite the presence of noisy pseudo-labels [9], [10].

To mitigate the reliance on massive annotations, recent research has made substantial strides in semi-supervised learning, evolving through three primary paradigms in the RS domain: generative adversarial networks (GANs) [11], [12], consistency-based methods [13], [14], [15], [16], [17], and pseudo-labeling methods [18], [19], [20], [21]. Early GAN-based approaches, exemplified by Hung et al. [11] and Souly et al. [12], align feature distributions via adversarial learning but often suffer from training instability and convergence difficulties. Subsequently, consistency-based methods like FixMatch [16] and UniMatch [17] became mainstream by enforcing prediction invariance under perturbations. However, they remain susceptible to confirmation bias, where incorrect predictions are reinforced over time. Similarly, pseudo-labeling strategies like U2PL [19] and DWL [22] expand the training set with high-confidence predictions, yet they struggle with noise accumulation when the initial predictions are unreliable. Crucially, since these paradigms primarily rely on self-generated supervisory signals to guide training, they remain vulnerable to pseudo-label drift, particularly in complex RS scenes. Without strong external guidance to rectify errors, these methods frequently encounter difficulties in distinguishing semantically similar categories or delineating precise object boundaries under severe label scarcity.

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To tackle these challenges, we propose Co2S, a stable semi-supervised RS segmentation framework. We construct a heterogeneous dual-student architecture that synergizes complementary priors derived from vision-language models (VLMs) and self-supervised models (SSMs) to mitigate pseudo-label drift. The framework is underpinned by an explicit–implicit semantic collaborative guidance mechanism and a global–local feature collaborative fusion strategy. Collectively, these components establish a stable mutual learning process, ensuring superior segmentation performance even under severe label scarcity.

The contributions of this work are summarized as follows.

1) We propose a stable dual-student learning framework for semi-supervised RS segmentation, instantiated with two heterogeneous ViT-based vision foundation models. By leveraging distinct priors from pretrained CLIP and DINOv3, it enables reliable mutual learning and significantly alleviates pseudo-label drift.
2) We introduce an explicit–implicit semantic co-guidance mechanism that utilizes text embeddings and learnable queries to provide explicit and implicit class-level guidance, thereby jointly enhancing semantic consistency.
3) We develop a global–local feature co-fusion strategy, effectively fusing the global contextual information captured by CLIP with the local texture details produced by DINOv3. This complementary fusion enables the model to generate highly precise segmentation results.

Extensive experiments on six diverse RS benchmarks, including WHDLD [23], LoveDA [24], Potsdam [25], GID-15 [26], MER [27], and MSL [27], demonstrate the effectiveness of Co2S. The results show that our framework consistently delivers leading performance across various partition protocols, outperforming existing semi-supervised methods, particularly in regimes with extremely scarce annotations.

II. Related Work

A. RS Semi-Supervised Semantic Segmentation

In the early exploration phase, GANs were incorporated into the semi-supervised segmentation framework to serve as an auxiliary supervision branch, facilitating model optimization through the adversarial discrimination of prediction distributions. Souly et al. [12] leveraged a generator network to produce fake images using class-level information as extra training examples, forcing real samples to cluster in the feature space to improve pixel classification. Focusing on the output space, Hung et al. [11] designed a discriminator in a fully convolutional manner to differentiate predicted probability maps from the ground truth, enabling the discovery of trustworthy regions in unlabeled data to facilitate effective semi-supervised learning.

Subsequently, consistency-based methods gained prominence by enforcing prediction invariance under perturbations. Specifically, FixMatch [16] utilizes high-confidence predictions from weakly augmented images to supervise strongly augmented ones, providing a simplified yet effective means to leverage unlabeled data. Building on this, UniMatch [17] introduces dual-stream input and feature perturbations to guide two strong views with a common weak view, successfully exploring a broader perturbation space for superior segmentation accuracy. Tailored for remote sensing, WSCL [28] employs strong data augmentation and adaptive reweighting to decouple self-biased predictions and alleviate overfitting. Similarly, MUCA [29] integrates multiscale uncertainty consistency and cross-teacher–student attention to construct discriminative feature representations, improving the distinction of highly similar objects. Parallel to consistency strategies, pseudo-labeling methods represent another active research direction, focusing on generating high-quality labels for self-training. U2PL [19] enhances data utilization by identifying unreliable pixels based on entropy and using them as negative samples. Addressing the specific challenges of noise and class imbalance in remote sensing, DWL [22] employs a decoupled learning module alongside a ranking-based weighting module to adaptively filter and balance pseudo-labels. Different from these methods that rely on homogeneous co-training architectures or standard teacher–student paradigms, Co2S introduces a heterogeneous dual-student paradigm. By exploiting the complementary nature of distinct vision foundation models, our framework breaks the confirmation bias inherent in conventional co-training, offering an effective solution for pseudo-label drift.

B. Semantic Segmentation With Vision Foundation Models

Vision foundation models have marked a paradigm shift in semantic segmentation, broadly branching into VLMs and SSMs. VLMs [30], [31], [32], [33], [34] leverage text–image alignment for transferable visual understanding. CLIP [30] utilizes large-scale contrastive pretraining to distill generic visual concepts, while methods like DenseCLIP [31] and MaskCLIP [32] adapt it for dense prediction tasks. In remote sensing, RemoteCLIP [33] aligns visual features of RS images with textual descriptions to learn robust visual features with rich semantics. Parallelly, SSMs [35], [36], [37], [38], [39], [40], [41], [42] focus on extracting intrinsic spatial features, capturing hierarchical visual patterns without relying on manual annotations or textual supervision. The DINO family [35], [36], [37] employs self-distillation to learn high-quality dense features effectively, whereas RingMo [38] leverages generative self-supervised learning to specifically target dense and small objects in RS scenes, Cloud-Adapter [43] proposes a parameter-efficient tuning strategy that adapts DINOv2 [36] to the cloud segmentation without retraining the backbone.

Building on these advancements, recent studies explore architectures incorporating both VLMs and SSMs components [44], [45], [46]. Barsellotti et al. [46] proposed Talk2DINO, which aligns CLIP's textual embeddings with DINO's visual features via a learned mapping, resulting in less noisy segmentations with improved foreground distinction. For remote sensing, Ye et al. [47] developed GSNet, employing query-guided feature fusion to integrate representations from generalist and specialist encoders to achieve more accurate mask predictions. Similarly, RSKT-Seg [44] aggregates multidirectional cost maps to capture rotation-invariant cues, thereby enhancing both inference efficiency and segmentation accuracy. Alternatively, shifting the focus

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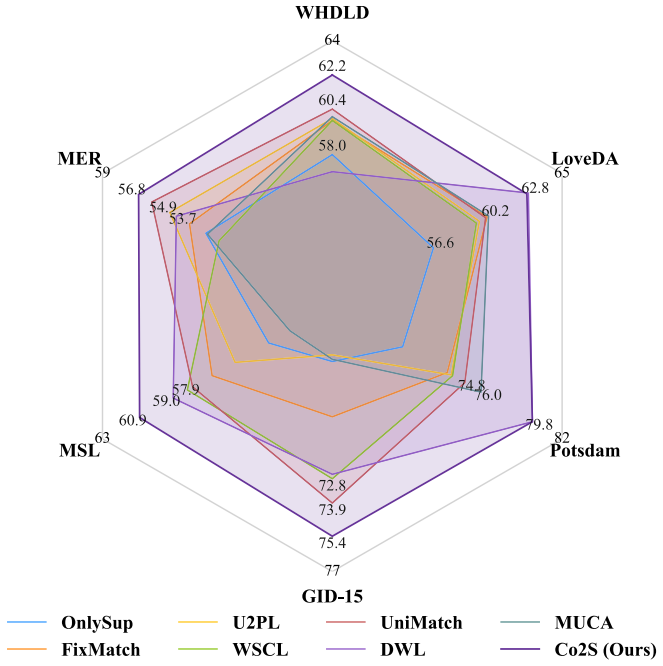


Fig. 1. Radar chart compares the mIoU (%) of different semi-supervised semantic segmentation methods across six RS datasets under the 1/8 labeled ratio. Co2S consistently maintains leading performance across all benchmarks.

to utilizing frozen representations, Zhang et al. [45] leveraged the static backbones of CLIP and DINO to generate high-quality pseudo-labels, enabling effective weakly supervised segmentation with reduced training costs. In a similar vein, CLIP-DINOiser [48] applies DINO's spatial affinities to refine the dense MaskCLIP features, producing smoother segmentation outputs without the need for pixel-level annotations. However, these approaches primarily target open-vocabulary supervised or weakly supervised scenarios. The potential of synergizing CLIP and DINO within a semi-supervised mutual learning framework, specifically to mitigate pseudo-label drift under label scarcity, remains unexplored. It is worth noting that adaptive modulation strategies have also shown promise in multistyle generative fusion [49], [50], further inspiring the exploration of synergistic feature interaction in dense prediction tasks. Our work bridges this gap by integrating their distinct priors to stabilize training when pixel-level annotations are severely limited.

III. Method

A. Stable Dual-Student Semi-Supervised Learning Framework

As illustrated in Fig. 2, Co2S adopts a dual-student semi-supervised segmentation pipeline instantiated with heterogeneous CLIP [30] and DINOv3 [37] encoders. The two students process the same augmented views in parallel and output segmentation logits. Specifically, the CLIP-based student is guided by explicit class-level semantics encoded from text, while the DINOv3-based student is driven by implicit class-level semantics represented by learnable queries. This design introduces category-indexed learnable queries that are optimized jointly with the segmentation objective, inspired by recent studies on modeling class-level representations in the visual feature space, such as D2LS [51] and Talk2DINO [46]. The two semantic representations are injected into isomorphic yet parameter-independent query-based decoders (see Section III-B). For labeled data, each student is supervised by the ground-truth mask. For unlabeled data, each student is trained with weak-to-strong consistency regularization across its own augmented predictions, while a global–local collaborative fusion mechanism is further introduced to perform confidence-driven pixel-wise mutual supervision between the two students (see Section III-C).

Formally, the training set consists of a labeled subset $\mathcal{D}_l = \{(\mathbf{x}_l, \mathbf{y}_l)\}$ and an unlabeled subset $\mathcal{D}_u = \{\mathbf{x}_u\}$, where $\mathbf{x}_l, \mathbf{x}_u \in \mathbb{R}^{3\times H\times W}$ and $\mathbf{y}_l \in \mathbb{R}^{1\times H\times W}$. Each student outputs segmentation logits with N classes. For a labeled sample $(\mathbf{x}_l, \mathbf{y}_l)$, we generate a weakly augmented view $\mathbf{x}_l^w$ and feed it into both students, obtaining $\mathbf{Z}^{\text{global}} \in \mathbb{R}^{N\times H\times W}$ from the CLIP-based student and $\mathbf{Z}^{\text{local}} \in \mathbb{R}^{N\times H\times W}$ from the DINOv3-based student. The supervised objective is applied independently

$$\mathcal{L}_{\text{sup}}^{\text{global}} = \text{CE}\left(\mathbf{Z}^{\text{global}}, \mathbf{y}_l\right), \quad \mathcal{L}_{\text{sup}}^{\text{local}} = \text{CE}\left(\mathbf{Z}^{\text{local}}, \mathbf{y}_l\right) \tag{1}$$

where $\text{CE}(\cdot,\cdot)$ denotes the pixel-wise cross entropy between logits and the ground-truth mask and $\mathcal{L}_{\text{sup}}^{\text{global}}$ and $\mathcal{L}_{\text{sup}}^{\text{local}}$ denote the supervised losses for the CLIP- and DINOv3-based students, respectively.

For an unlabeled sample $\mathbf{x}_u$, we strictly adhere to the dual-stream perturbation paradigm of UniMatch [17]. We generate a weakly augmented view $\mathbf{x}_u^w$, two strongly augmented views $\mathbf{x}_u^{s1}, \mathbf{x}_u^{s2}$, and one feature-perturbed view $\mathbf{x}_u^{fp}$, where $\mathbf{x}_u^w, \mathbf{x}_u^{s1}, \mathbf{x}_u^{s2}, \mathbf{x}_u^{fp} \in \mathbb{R}^{3\times H\times W}$. For each student, let $\mathbf{Z}^w \in \mathbb{R}^{N\times H\times W}$ denote the segmentation logits predicted from the weak view and $\mathbf{p}^w = \text{Softmax}(\mathbf{Z}^w)$ denote the probability map, where $\text{Softmax}(\cdot)$ is applied over the class dimension. For each pixel at location (i, j), we assign the pseudo-label $\hat{y}_{u,i,j} = \arg\max_n(\mathbf{p}_{n,i,j}^w)$ and the confidence score $c_{i,j} = \max_n(\mathbf{p}_{n,i,j}^w)$, where $\hat{y}_{u,i,j} \in \{1,\ldots,N\}$ and $c_{i,j} \in [0,1]$. To formulate the consistency regularization, we define a generic masked cross-entropy function $\mathcal{H}$ given a logit map $\mathbf{Z} \in \mathbb{R}^{N\times H\times W}$, a pseudo-label map $\hat{\mathbf{y}}_u \in \mathbb{R}^{1\times H\times W}$, and confidence scores $\mathbf{c} \in \mathbb{R}^{1\times H\times W}$

$$\mathcal{H}\left(\mathbf{Z}, \hat{\mathbf{y}}_u, \mathbf{c}\right) = \sum_{i,j} \mathbf{1}\left(c_{i,j} \geq \tau\right) \cdot \text{CE}\left(\mathbf{Z}_{:,i,j}, \hat{y}_{u,i,j}\right) \tag{2}$$

where $\mathbf{1}(\cdot)$ is the indicator function and τ is the confidence threshold. The final consistency loss $\mathcal{L}_{\text{ct}}$ is calculated as

$$\mathcal{L}_{\text{ct}} = \frac{1}{2}\mathcal{H}\left(\mathbf{Z}^{fp}, \hat{\mathbf{y}}_u, \mathbf{c}\right) + \frac{1}{4}\left(\mathcal{H}\left(\mathbf{Z}^{s1}, \hat{\mathbf{y}}_u, \mathbf{c}\right) + \mathcal{H}\left(\mathbf{Z}^{s2}, \hat{\mathbf{y}}_u, \mathbf{c}\right)\right) \tag{3}$$

where $\mathbf{Z}^{fp}, \mathbf{Z}^{s1}, \mathbf{Z}^{s2} \in \mathbb{R}^{N\times H\times W}$ are the logits predicted from the feature-perturbed view and the two strongly augmented views, respectively.

For each student, the total training objective is formulated as the weighted sum of the supervised, consistency, and stability losses

$$\mathcal{L} = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{ct}}\mathcal{L}_{\text{ct}} + \lambda_{\text{sta}}\mathcal{L}_{\text{sta}}. \tag{4}$$

Here, $\mathcal{L}_{\text{sup}}$ refers to $\mathcal{L}_{\text{sup}}^{\text{global}}$ for the CLIP-based student and $\mathcal{L}_{\text{sup}}^{\text{local}}$ for the DINOv3-based student, and $\mathcal{L}_{\text{ct}}$ and $\mathcal{L}_{\text{sta}}$ are

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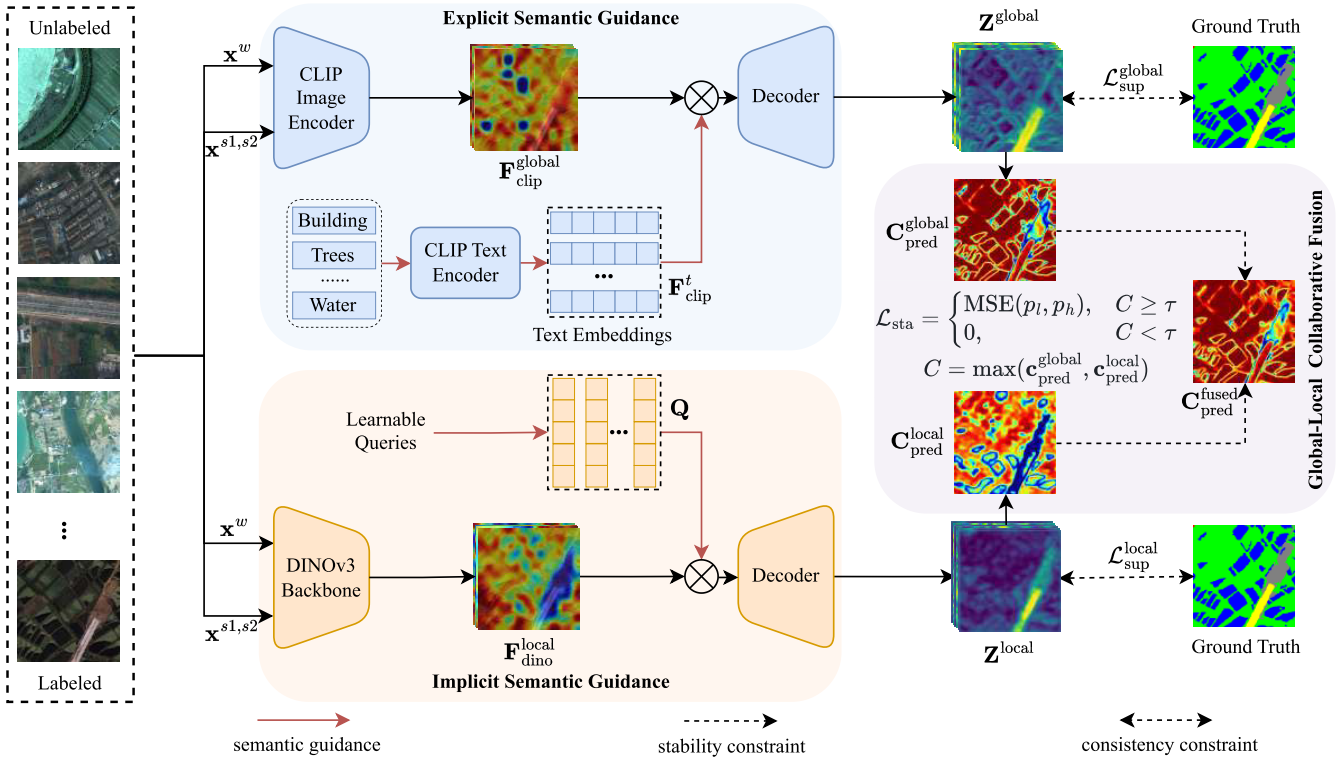


Fig. 2. Overview of the proposed Co2S framework. (Top) It integrates a CLIP-based student using text embeddings for explicit semantic guidance and a DINOv3-based student (bottom) using learnable queries for implicit guidance. For unlabeled data, the global–local collaborative fusion strategy enforces training stability by arbitrating supervision based on pixel-wise confidence.

defined analogously. Following the standard practice in UniMatch [17], we set $\lambda_{\text{sup}} = \lambda_{\text{ct}} = 0.5$ and use a cosine ramp-up schedule for λ_{sta} that increases from 0 to 0.5 over the first 20% of the total training steps, after which λ_{sta} remains fixed at 0.5. The detailed definition of stability loss $\mathcal{L}_{\text{sta}}$ is provided in Section III-C.

B. Explicit–Implicit Semantic Collaborative Guidance

To fully exploit heterogeneous pretraining, Co2S injects complementary semantic priors into the two students. The CLIP-based student receives explicit class-level guidance from a frozen text encoder, while the DINOv3-based student is driven by implicit class-level guidance via learnable queries.

For the CLIP-based student, we adopt a concept-based prompt ensembling strategy to improve semantic precision. For each semantic category $k \in \{1, \ldots, N\}$, we manually curate a set of fine-grained concept descriptions $\mathcal{C}_k = \{c_{k,i}\}_{i=1}^{S_k}$, typically comprising 5–6 distinct items per class to provide diverse semantic descriptions for each category. A representative example of the concept sets used for the Potsdam dataset is detailed in Table I. Each concept $c_{k,i} \in \mathcal{C}_k$ is formatted by a prompt template function $\mathcal{T}(\cdot)$ (e.g., $\mathcal{T}(c_{k,i})$ = "a photo of a {concept}") and then encoded by the frozen CLIP text encoder $\mathcal{E}_{\text{txt}}$. We aggregate these concept embeddings by averaging to obtain a class prototype $\mathbf{t}_k \in \mathbb{R}^{C_1}$

$$\mathbf{t}_k = \frac{1}{S_k} \sum_{i=1}^{S_k} \mathcal{E}_{\text{txt}} \left(\mathcal{T} \left(c_{k,i} \right) \right). \tag{5}$$

TABLE I
LIST OF FINE-GRAINED CONCEPT DESCRIPTIONS MANUALLY CURATED FOR EACH CATEGORY IN THE POTSDAM DATASET

Category	Concept Descriptions
Impervious surface	impervious surface, asphalt, concrete, pavement, roof, sidewalk
Building	building, residential building, commercial building, church, factory, historical building
Low vegetation	low vegetation, grass, lawn, meadow, pasture, flower bed
Tree	tree, deciduous tree, evergreen tree, tree row, park tree, forest edge
Car	car, passenger car, truck, van, bus, parked vehicle
Clutter	clutter, fence, wall, railway, construction site, unclassified object

Finally, we stack the prototypes of all N categories into an explicit text-query matrix $\mathbf{F}_{\text{clip}}^{t} \in \mathbb{R}^{N \times C_1}$

$$\mathbf{F}_{\text{clip}}^{t} = [\mathbf{t}_1, \ldots, \mathbf{t}_N] \tag{6}$$

which provides explicit class-level guidance for all categories using language-derived semantics.

In contrast, for the DINOv3-based student, we utilize learnable queries $\mathbf{Q} \in \mathbb{R}^{N \times C_2}$ to provide implicit class-level

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guidance [51], [52]. Specifically, $\mathbf{Q}$ contains N learnable query vectors, where each query corresponds to one fixed category slot

$$\mathbf{Q} = \left[\mathbf{q}^1, \ldots, \mathbf{q}^N\right]. \tag{7}$$

These queries are implemented as a trainable embedding layer and optimized jointly with the DINOv3 backbone and the segmentation decoder. Although randomly initialized, each query is consistently aligned with one prediction channel and gradually evolves into a category-associated representation under category-aligned pixel-wise supervision.

Let $\mathbf{F}_{\text{clip}}^{\text{global}} \in \mathbb{R}^{C_1 \times h \times w}$ and $\mathbf{F}_{\text{dino}}^{\text{local}} \in \mathbb{R}^{C_2 \times h \times w}$ denote the visual features from the CLIP-based and DINOv3-based students, respectively. Both students employ an isomorphic yet parameter-independent query-based decoder, and compute segmentation logits by dot-product similarity

$$\mathbf{Z}^{\text{global}} = \text{Sim}\left(\mathbf{F}_{\text{clip}}^{t}, \mathbf{F}_{\text{clip}}^{\text{global}}\right), \quad \mathbf{Z}^{\text{local}} = \text{Sim}\left(\mathbf{Q}, \mathbf{F}_{\text{dino}}^{\text{local}}\right) \tag{8}$$

where $\mathbf{Z}^{\text{global}}, \mathbf{Z}^{\text{local}} \in \mathbb{R}^{N \times h \times w}$ and $\text{Sim}(\cdot,\cdot)$ denotes a dot-product similarity implemented with an `einsum` operator. Accordingly, the category information carried by $\mathbf{Q}$ arises not from manual semantic initialization, but from its category-indexed parameterization and category-aligned optimization. This formulation maintains a unified architectural structure while allowing the two students to leverage distinct semantic sources, namely explicit text embeddings versus implicit learnable queries.

Although the CLIP text embeddings $\mathbf{F}_{\text{clip}}^{t}$ and the DINOv3 learnable queries $\mathbf{Q}$ are independent in the forward pass, they become coupled during optimization via the stability loss $\mathcal{L}_{\text{sta}}$. On mutually confident pixels, $\mathcal{L}_{\text{sta}}$ enforces consistency between the two predictions. Crucially, this mechanism enables the explicit class-level guidance from CLIP to rectify potential semantic category confusion in the DINOv3-based student. Simultaneously, the implicit class-level guidance from DINOv3 reciprocally refines the coarse spatial localization of the CLIP-based student. Through this explicit and implicit semantic collaboration, Co2S effectively mitigates pseudo-label drift and stabilizes the dual-student learning process.

C. Global–Local Feature Collaborative Fusion

Conventional co-training frameworks [14], [15], [53] typically rely on identical network architectures with different random initializations. Despite the divergent starting points, these homogeneous models often converge to similar error patterns due to confirmation bias. To mitigate this error accumulation, Co2S introduces a global–local feature collaborative fusion strategy. As visually evidenced in Fig. 3, the attention maps reveal distinct representational characteristics where CLIP produces diffuse activations rich in global contextual information while DINOv3 generates sharp responses highlighting local details. These distinct behaviors are mainly induced by different pretraining objectives rather than the shared Transformer backbone. CLIP is trained with image–text contrastive alignment to capture language-aligned semantics and global context, while DINOv3 is trained with self-supervised distillation to learn fine-grained patch correspondence and local structures. Similar observations have also been reported in recent studies [46], [48]. Building on this complementarity, our mechanism effectively fuses the global contextual information captured by CLIP [30] with the local details produced by DINOv3 [37]. This complementary fusion enables the model to generate segmentation maps with more accurate semantics and sharper, well-defined boundaries.

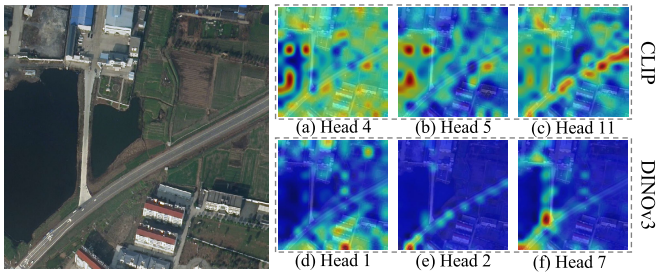


Fig. 3. Visualization of attention maps from different heads of (a)–(c) CLIP image encoder and (d)–(f) DINOv3 backbone.

Given an unlabeled image $\mathbf{x}_u$, its weakly augmented view is fed into the two students to obtain the global logits $\mathbf{Z}^{\text{global}} \in \mathbb{R}^{N \times H \times W}$ and the local logits $\mathbf{Z}^{\text{local}} \in \mathbb{R}^{N \times H \times W}$. We first compute the confidence maps $\mathbf{C}_{\text{pred}}^{\text{global}} \in \mathbb{R}^{1 \times H \times W}$ and $\mathbf{C}_{\text{pred}}^{\text{local}} \in \mathbb{R}^{1 \times H \times W}$

$$\mathbf{C}_{\text{pred}}^{\text{global}} = \max_n \left[\text{Softmax}\left(\mathbf{Z}^{\text{global}}\right)\right]_n \tag{9}$$

$$\mathbf{C}_{\text{pred}}^{\text{local}} = \max_n \left[\text{Softmax}\left(\mathbf{Z}^{\text{local}}\right)\right]_n \tag{10}$$

where $\text{Softmax}(\cdot)$ is applied over the class dimension. Let $c_{\text{pred}}^{\text{global}}$ and $c_{\text{pred}}^{\text{local}}$ denote the scalar confidence values at a specific spatial location from $\mathbf{C}_{\text{pred}}^{\text{global}}$ and $\mathbf{C}_{\text{pred}}^{\text{local}}$, respectively. The joint maximum confidence for this pixel is defined as

$$C = \max\left(c_{\text{pred}}^{\text{global}}, c_{\text{pred}}^{\text{local}}\right). \tag{11}$$

We then build interaction masks based on three confidence scenarios.

1) If $c_{\text{pred}}^{\text{global}} \geq \tau$ and $c_{\text{pred}}^{\text{local}} \geq \tau$, the lower confidence student learns from the higher confidence one; when their confidence is equal, they engage in symmetric mutual learning.

2) If $C \geq \tau$ but one of $\{c_{\text{pred}}^{\text{global}}, c_{\text{pred}}^{\text{local}}\}$ is below τ, the low-confidence student is supervised by the high-confidence peer.

3) If $C < \tau$, the pixel is ignored and does not contribute to stabilization. Formally, the stability loss is defined as

$$\mathcal{L}_{\text{sta}}(t) = \lambda(t) \begin{cases} \text{mse}\left(\mathbf{p}_\ell, \mathbf{p}_h\right), & C \geq \tau \\ 0, & C < \tau \end{cases} \tag{12}$$

where $\text{mse}(\cdot,\cdot)$ represents the mean squared error calculation and $\mathbf{p}_\ell, \mathbf{p}_h \in \mathbb{R}^N$ denote the class-probability vectors of the lower and higher confidence students at each pixel, respectively. They are computed by applying $\text{Softmax}(\cdot)$ to the corresponding logits, τ is the confidence threshold, and $\lambda(t)$ is a ramp-up weight depending on the training step t. In Fig. 2, we use the notation $\mathbf{C}_{\text{pred}}^{\text{fused}} \in \mathbb{R}^{1 \times H \times W}$ to represent the image-level map of the joint maximum confidence values, whose entries correspond to the per-pixel scalar C in (11).

This pixel-wise fusion strategy bridges global contextual clues with local texture details based on pixel-wise confidence. By synergizing these distinct priors at the optimization

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level, the proposed strategy effectively compensates for limited supervision, yielding predictions that are both semantically stable and spatially precise even under scarce labels.

IV. Experiments

A. Datasets

We evaluate Co2S on six widely used RS segmentation benchmarks, covering a diverse range of sensors, spatial resolutions, geographic regions, and scene complexities.

1) WHDLD: WHDLD [23] comprises 4940 optical images captured by GaoFen-1 and ZiYuan-3 satellites, each having a size of 256 × 256 pixels and a 2-m spatial resolution. It provides pixel-wise annotations for six categories: building, road, pavement, vegetation, bare soil, and water.

2) LoveDA: LoveDA [24] provides 5987 images of size 1024 × 1024 pixels with 0.3-m resolution. Spanning urban and rural scenes, it annotates seven categories: background, building, road, water, barren, forest, and agriculture.

3) Potsdam: Potsdam [25] consists of 38 aerial orthophotos with 5-cm resolution and 6000 × 6000 dimensions. It covers six classes: impervious surface, building, low vegetation, tree, car, and clutter. These images are cropped into 5472 nonoverlapping 512 × 512 patches for training and evaluation.

4) GID-15: GID-15 [26] originates from ten GaoFen-2 satellite images with approximate dimensions of 7200 × 6800 pixels. It provides pixel-level annotations for 15 fine-grained categories, spanning industrial, residential, agricultural, and water regions. These large images are tiled into 3420 nonoverlapping 512 × 512 patches to facilitate large-scale land-cover classification.

5) MER and MSL: Derived from HiRISE imagery for Mars terrain analysis [27], MER contains 1023 grayscale images sized 1024 × 1024, while MSL comprises 4155 RGB images measuring 560 × 500. They share nine distinct planetary surface categories: martian soil, sands, gravel, bedrock, rocks, tracks, shadows, unknown, and background.

B. Implementation Details

The proposed Co2S framework is optimized using the AdamW optimizer [54] with a weight decay of 0.01 and a polynomial learning rate decay schedule (power = 0.9). Structurally, both students utilize a ViT-B/16 backbone [55] and employ decoders with identical architectures but independent parameters. Regarding data augmentation, we strictly follow the protocols established in UniMatch [17]. Specifically, labeled images and "weak" unlabeled views undergo basic geometric transformations, including random resizing (scale range 0.5-2.0) and horizontal flipping. "strong" unlabeled views are generated via a strong augmentation pipeline comprising color jittering, random grayscale conversion, Gaussian blur, and CutMix [56]. Furthermore, feature-space perturbation is applied via dropout [57] (rate = 0.5) within the dual-student forward mechanism. The unsupervised loss is guided by confidence-thresholded pseudo-labels (threshold = 0.95). Regarding the training schedule, each experiment runs for 80 epochs, with the exception of the MER dataset [27]. Due to its limited sample size, MER requires a prolonged schedule of 150 epochs to mitigate noisy updates and ensure stable convergence. All experiments were conducted on a workstation equipped with eight NVIDIA RTX 3090 GPUs. We utilize the mean intersection over union (mIoU) as the evaluation metric. During inference, we fix the CLIP-based student as the prediction model, as it consistently achieves higher mIoU than the DINOv3-based student on the validation set across all label ratios. This design ensures identical inference cost to single-model baselines without ensemble overhead.

TABLE II

Segmentation Results (mIoU %) on the WHDLD Dataset Under Different Labeled Ratios. The Best and Second-Best Results Are Highlighted in Bold and Underlined, Respectively

Method	1/24	1/16	1/8	1/4
OnlySup	53.6	55.2	58.0	60.5
FixMatch [16] [NeurIPS'20]	56.0	57.6	59.8	60.8
U2PL [19] [CVPR'22]	57.1	58.3	59.9	61.1
WSCL [28] [TGRS'23]	56.8	58.6	59.8	61.1
UniMatch [17] [CVPR'23]	<u>57.4</u>	<u>58.8</u>	<u>60.4</u>	<u>61.5</u>
DWL [22] [ISPRS'24]	56.5	57.0	57.1	58.9
MUCA [29] [TGRS'25]	56.5	58.2	60.0	60.5
Co2S (Ours)	**61.1**	**61.5**	**62.2**	**62.6**

TABLE III

Segmentation Results (mIoU %) on the LoveDA Dataset Under Different Labeled Ratios. The Best and Second-Best Results Are Highlighted in Bold and Underlined, Respectively

Method	1/40	1/16	1/8	1/4
OnlySup	45.9	53.1	56.6	58.9
FixMatch [16] [NeurIPS'20]	53.3	57.9	60.1	61.5
U2PL [19] [CVPR'22]	52.3	58.0	59.6	61.5
WSCL [28] [TGRS'23]	54.1	57.4	59.4	61.1
UniMatch [17] [CVPR'23]	55.5	58.6	60.0	61.7
DWL [22] [ISPRS'24]	<u>57.4</u>	<u>59.0</u>	**62.8**	<u>63.4</u>
MUCA [29] [TGRS'25]	54.6	56.7	60.2	63.1
Co2S (Ours)	**58.2**	**60.4**	<u>62.7</u>	**64.0**

C. Comparison With State-of-the-Art Methods

We report comparison results across six widely used remote sensing benchmarks to evaluate the performance of the proposed Co2S comprehensively. We conduct extensive comparisons against the supervised baseline ("OnlySup"), which is trained exclusively on the labeled data, as well as several general semi-supervised baselines, including FixMatch [16], UniMatch [17], U2PL [19], and remote sensing-specific approaches like WSCL [28], DWL [22], and MUCA [29].

1) Quantitative Comparison and Analysis: The comprehensive comparisons across six diverse benchmarks (Tables II–VI) demonstrate that Co2S consistently achieves leading performance, particularly in regimes with extremely scarce annotations. On the WHDLD [23] dataset, Co2S outperforms the strong competitor UniMatch by 3.7% under the

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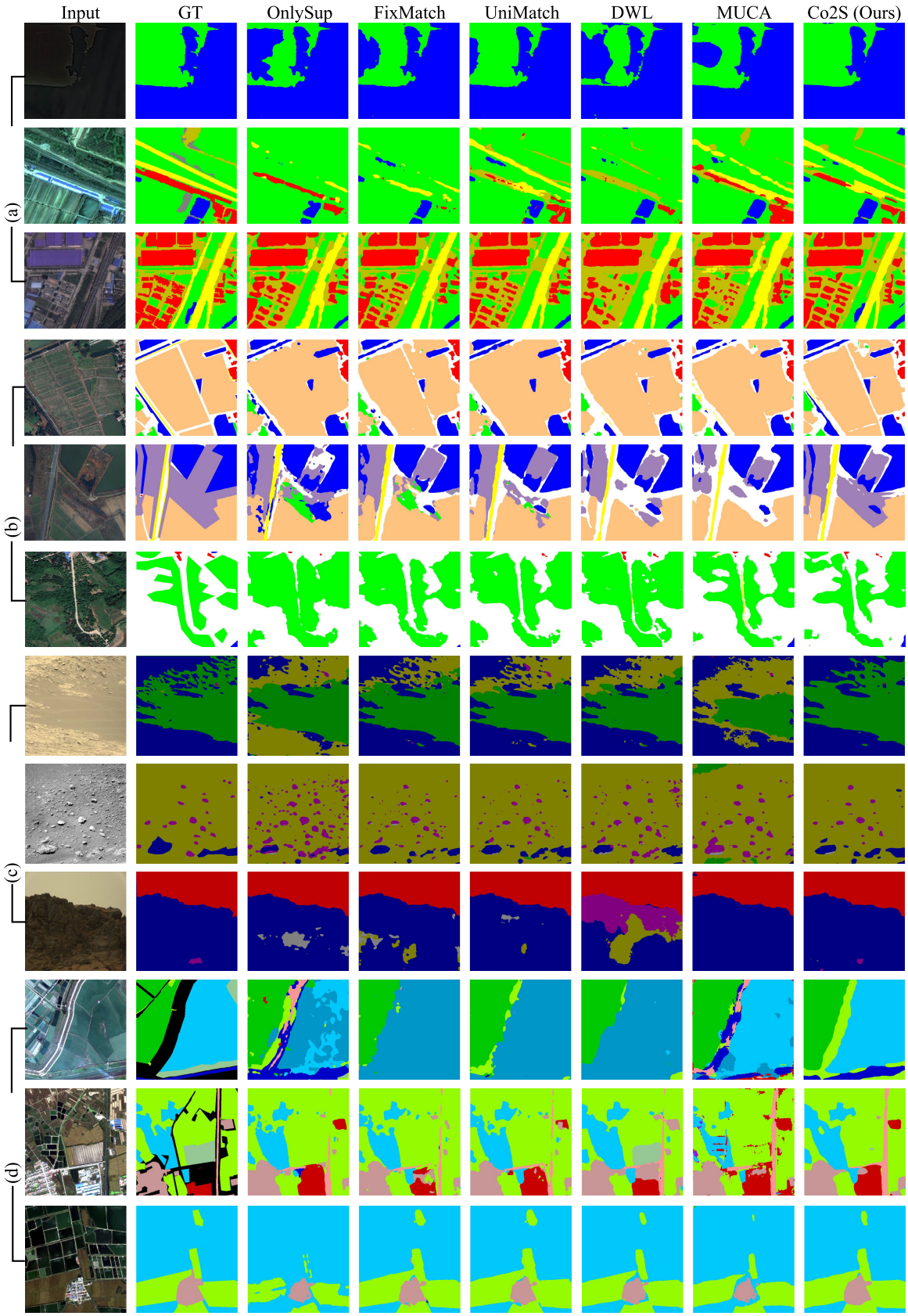


Fig. 4. Visual comparison of semantic segmentation results on the five datasets under the 1/8 labeled ratio. (a) WHDLD. (b) LoveDA. (c) MER and MSL. (d) GID-15.

extremely sparse 1/24 split, demonstrating its capability to parse diverse land-cover features accurately. Similarly, on the LoveDA [24] dataset, the method yields a remarkable 12.3% improvement over the supervised baseline at the 1/40 ratio,

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TABLE IV
SEGMENTATION RESULTS (MIOU %) ON THE POTSDAM DATASET UNDER DIFFERENT LABELED RATIOS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	1/32	1/16	1/8	1/4
OnlySup	61.5	64.9	70.2	73.8
FixMatch [16] [NeurIPS'20]	69.4	71.5	73.5	75.7
U2PL [19] [CVPR'22]	67.2	71.3	73.8	75.9
WSCL [28] [TGRS'23]	69.9	72.3	73.9	75.4
UniMatch [17] [CVPR'23]	70.7	72.6	74.8	76.4
DWL [22] [ISPRS'24]	<u>74.2</u>	**78.3**	**79.8**	**80.3**
MUCA [29] [TGRS'25]	66.9	71.5	<u>76.0</u>	79.2
Co2S (Ours)	**74.3**	<u>76.6</u>	**79.8**	<u>80.2</u>

TABLE V
SEGMENTATION RESULTS (MIOU %) ON THE MER AND MSL DATASETS UNDER DIFFERENT LABELED RATIOS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	MER		MSL	
	1/8	1/4	1/8	1/4
OnlySup	52.8	55.3	53.6	59.0
FixMatch [16] [NeurIPS'20]	53.8	<u>58.1</u>	56.8	60.5
U2PL [19] [CVPR'22]	54.9	56.1	55.5	61.6
WSCL [28] [TGRS'23]	51.9	54.9	58.2	59.9
UniMatch [17] [CVPR'23]	<u>56.0</u>	<u>58.1</u>	57.9	62.8
DWL [22] [ISPRS'24]	54.5	54.7	<u>59.0</u>	<u>63.6</u>
MUCA [29] [TGRS'25]	52.6	53.0	52.4	59.3
Co2S (Ours)	**56.8**	**59.1**	**60.9**	**65.9**

showing superior performance despite significant urban–rural distribution shifts. For the ultrahigh-resolution Potsdam [25] dataset, Co2S achieves the best performance at the 1/32 split with 74.3% mIoU, surpassing remote sensing-specific designs like DWL. Extending to the large-scale GID-15 [26] dataset, our method outperforms UniMatch by 1.5% at the 1/8 split, indicating proficiency in handling diverse land-cover characteristics across vast spatial areas. Furthermore, in the extraterrestrial environments of MER and MSL [27], Co2S delivers the highest accuracy on these benchmarks characterized by extreme class imbalance. Crucially, a consistent trend observed across all benchmarks is that the performance advantage of Co2S over baselines amplifies as the labeled ratio decreases. This phenomenon validates the stability of our Co2S, confirming that the synergistic foundation model priors effectively prevent model collapse and sustain high accuracy even in the most extreme low-data annotation regimes.

2) Qualitative Visualization Analysis: To further substantiate these quantitative findings, we provide visual comparisons of segmentation results across five datasets in Fig. 4. As observed, baseline methods frequently struggle with semantic confusion in complex scenes. For instance, in the WHDLD dataset [Fig. 4(a)], baseline methods incorrectly classify large water regions (blue) as vegetation (green). Similarly, in LoveDA [Fig. 4(b)], competing approaches exhibit notable category confusion, where barren land (purple) is misclassified as either background (white) or forest (green). Moreover, our method demonstrates superior capability in handling fine-grained objects and structural details. In the MER and MSL datasets [Fig. 4(c)], baseline methods suffer from severe over-segmentation, generating excessive false positives for small rocks (purple), whereas Co2S effectively suppresses such noise and produces results that better match the ground truth. Likewise, in GID-15 [Row Fig. 4(d)], baseline methods frequently produce discontinuous predictions for thin traffic land (pink) and fail to correctly identify irrigated land (light green), whereas Co2S yields more complete and consistent segmentation results. These visualizations confirm that our framework effectively harmonizes global semantic consistency with local structural precision across diverse and challenging RS environments, successfully rectifying category confusion while preserving fine-grained details.

3) Pseudo-Label Quality and Stability: To empirically verify our claim that Co2S effectively mitigates pseudo-label drift, we monitor the quality of pseudo-labels during the critical early training phase. The pseudo-label accuracy is calculated as the proportion of correctly classified pixels among those that pass the confidence threshold τ

$$\text{Accuracy} = \frac{\sum \mathbf{1}\left(\hat{\mathbf{y}}_u = \mathbf{y}_{gt}\right) \cdot \mathbf{1}\left(c \geq \tau\right)}{\sum \mathbf{1}\left(c \geq \tau\right)} \tag{13}$$

where $\hat{\mathbf{y}}_u$ is the pseudo-label, $\mathbf{y}_{gt}$ is the ground truth, and c is the confidence score. The evolution curves for the first ten epochs on the WHDLD dataset (1/24 split) are plotted in Fig. 5. The results reveal distinct training dynamics as follows.

1) The supervised baseline (OnlySup) exhibits extreme volatility, oscillating drastically between 50% and 90%. This highlights the instability of learning from scratch with scarce annotations.

2) Standard methods like FixMatch [16] and UniMatch [17] show signs of early saturation. They quickly plateau at a suboptimal accuracy level (~80%–88%), suggesting that they are trapped by confirmation bias early in the training process and fail to correct initial errors.

3) In contrast, Co2S (red curve) demonstrates immediate semantic alignment. It surges to over 95% accuracy within the very first epoch and maintains this superior level with minimal variance. This rapid convergence and sustained stability (~10% gap over baselines) indicate that Co2S successfully establishes a high-quality supervision signal from the onset, thereby effectively stifling pseudo-label drift at its source and ensuring reliable optimization.

4) Arbitration Reliability and Confidence Calibration: To examine the arbitration assumption that higher confidence implies higher correctness, we analyze the confidence–accuracy correlation on labeled data. As shown in Fig. 6, empirical accuracy tends to increase with prediction confidence, indicating a clear positive correlation between confidence and correctness. In particular, predictions with confidence above the pseudo-label threshold ($\tau = 0.95$) achieve high empirical accuracy, suggesting that highly

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TABLE VI

SEGMENTATION RESULTS (MIOU %) ON THE GID-15 DATASET UNDER DIFFERENT LABELED RATIOS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	OnlySup	FixMatch [16] [NeurIPS'20]	U2PL [19] [CVPR'22]	WSCL [28] [TGRS'23]	UniMatch [17] [CVPR'23]	DWL [22] [ISPRS'24]	MUCA [29] [TGRS'25]	**Co2S (Ours)**
1/8	67.5	70.0	67.2	72.8	<u>73.9</u>	72.6	67.4	**75.4**
1/4	74.1	74.8	75.3	75.4	75.9	<u>76.9</u>	72.6	**77.7**

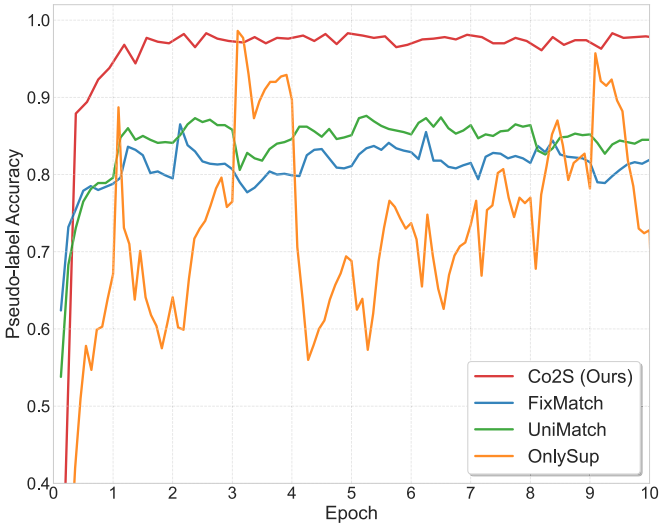


Fig. 5. Evolution of pseudo-label accuracy during the early training phase (first ten epochs) on the WHDLD dataset (1/24).

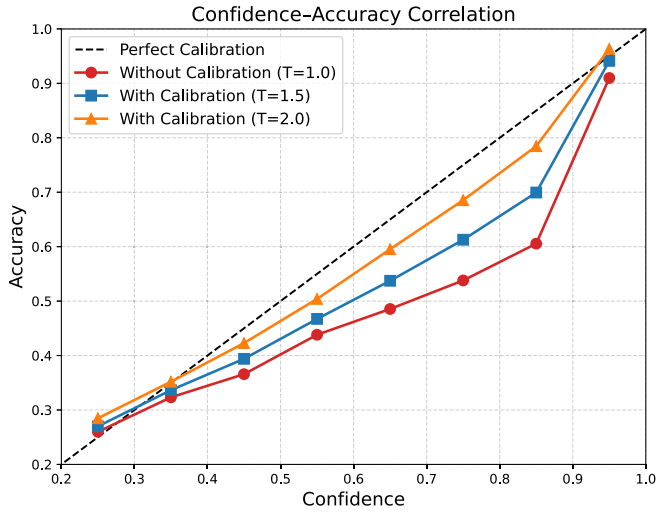


Fig. 6. Confidence–accuracy correlation on the WHDLD dataset (1/24) under different temperature scaling settings.

confident predictions are generally reliable for pseudo-label selection.

To further assess the impact of confidence calibration, we apply temperature scaling following [58]. Table VII reports the arbitration performance when the two students produce different predictions. Across all disagreement pixels, the pick-correct rate remains consistently above 60%, exceeding random selection (50%) and showing only minor variation across temperature settings. When restricted to the high-confidence region ($\tau \geq 0.95$), the arbitration accuracy reaches 80.38% under the uncalibrated setting ($T = 1.0$). Although

TABLE VII

EFFECT OF TEMPERATURE SCALING ON THE RELIABILITY OF MAX-CONFIDENCE ARBITRATION ON THE WHDLD DATASET (1/24). THE PICK-CORRECT RATE IS MEASURED ON ALL DISAGREEMENT PIXELS, WHILE THE ARBITRATION ACCURACY FOR $\tau \geq 0.95$ REFLECTS THE RELIABILITY OF HIGH-CONFIDENCE PSEUDO-LABEL SELECTION IN CO2S

T	Pick-Correct (%)	Arbitration Acc. ($\tau \geq 0.95$)	Pixels (M)
1.0	61.22	80.38	3.17
1.5	61.96	88.54	2.03
2.0	62.46	93.28	1.36

TABLE VIII

ABLATION STUDY ON THE EXPLICIT AND IMPLICIT SEMANTIC GUIDANCE MECHANISMS (ESG: EXPLICIT SEMANTIC GUIDANCE; ISG: IMPLICIT SEMANTIC GUIDANCE) ON THE WHDLD DATASET

ESG	ISG	mIoU
×	×	58.97
×	✓	58.86
✓	×	60.77
✓	✓	**61.09**

TABLE IX

ABLATION STUDY ON THE HETEROGENEITY OF THE DUAL-STUDENT ARCHITECTURE ON THE WHDLD DATASET (1/24)

Dual-student Architecture	mIoU
DINOv3 + DINOv3 (Local + Local)	45.20
CLIP + CLIP (Global + Global)	60.78
CLIP + DINOv3 (Global + Local)	**61.09**

TABLE X

ABLATION STUDY ON THE CONTRIBUTION OF DIFFERENT OPTIMIZATION OBJECTIVES ON THE WHDLD DATASET (1/24)

$\mathcal{L}_{sup}$	$\mathcal{L}_{ct}$	$\mathcal{L}_{sta}$	mIoU
✓	×	×	59.98
✓	✓	×	60.73
✓	✓	✓	**61.09**

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TABLE XI

COMPARISON OF DIFFERENT DIVERGENCE FORMULATIONS FOR THE STABILITY LOSS $\mathcal{L}_{\text{STA}}$ ON THE WHDLD DATASET (1/24). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RUNS

Divergence (for $\mathcal{L}_{\text{sta}}$)	mIoU (%)
KL ($T = 1$)	60.88 ± 0.07
JS ($T = 1$)	60.84 ± 0.06
Symmetric KL ($T = 1$)	60.82 ± 0.17
KL ($T = 2$)	60.95 ± 0.09
JS ($T = 2$)	60.82 ± 0.12
MSE	**61.04 ± 0.06**

TABLE XII

SENSITIVITY ANALYSIS OF THE MAXIMUM STABILITY LOSS WEIGHT $\lambda_{\text{STA}}^{\max}$ ON THE WHDLD DATASET (1/24)

$\lambda_{\text{sta}}^{\max}$	0.1	0.3	0.5	0.7
mIoU	60.93	60.90	**61.09**	60.85

TABLE XIII

SENSITIVITY ANALYSIS OF THE PSEUDO-LABEL CONFIDENCE THRESHOLD τ ON THE WHDLD DATASET (1/24). COV. DENOTES PSEUDO-LABEL COVERAGE AND P-ACC. DENOTES PSEUDO-LABEL ACCURACY

τ	Cov. (%)	P-Acc. (%)	mIoU
0.70	92.60	89.96	61.44
0.80	86.82	92.80	61.27
0.90	80.38	95.15	61.32
0.95	72.91	96.82	61.09
0.99	56.63	99.37	60.64

temperature scaling further improves arbitration accuracy, it substantially reduces the number of pixels whose confidence exceeds the pseudo-label threshold. Considering this tradeoff between reliability and sample availability, Co2S adopts the uncalibrated setting ($T = 1.0$) in practice. This observation is consistent with the sensitivity analysis of the confidence threshold τ, where increasing τ improves pseudo-label accuracy while reducing coverage.

D. Ablation Studies and Analysis

1) Impact of Explicit and Implicit Semantic Guidance: To validate the efficacy of the proposed explicit–implicit semantic co-guidance mechanism, we analyze the individual contributions of text embeddings and learnable queries, which provide explicit and implicit class-level guidance, respectively. As detailed in Table VIII, the baseline model without guidance achieves 58.97 mIoU. Implementing implicit class-level guidance via learnable queries $\mathbf{Q}$ alone results in a slight performance drop to 58.86. This decline is attributed to the fact that the learnable queries are randomly initialized parameters lacking pretrained priors. Optimizing these queries from scratch typically requires substantial annotated data, whereas the scarce labeled samples in our semi-supervised setting fail to provide sufficient supervision for their effective convergence. In contrast, integrating explicit class-level guidance via text embeddings $\mathbf{F}_{clip}^{t}$ alone yields a substantial gain of 1.8 and reaches 60.77 mIoU. This improvement stems from the inherent text–image alignment capability of CLIP acquired during pretraining, which enables the text embeddings to serve as powerful and reliable semantic anchors immediately. Notably, the optimal performance of 61.09 is attained only when both mechanisms are engaged. This result indicates a complementary synergy in which text embeddings ensure semantic correctness while the learnable queries adapt to dataset-specific visual patterns during training, thereby jointly enhancing semantic consistency.

2) Benefit From Global and Local Feature Fusion: To verify the effectiveness of the proposed global–local feature collaborative fusion strategy, we investigate the necessity of the heterogeneous dual-student design by comparing it against homogeneous initialization strategies, as detailed in Table IX. First, the homogeneous configuration where both students utilize DINOv3 performs poorly, yielding an mIoU of only 45.20. This indicates that without strong global semantic priors, students initialized solely with local structural features struggle to maintain category consistency, leading to severe semantic drift. Second, the homogeneous setting where both students employ CLIP attains a competitive mIoU of 60.78 due to rich global semantic representations. However, identical initialization results in highly correlated predictions. This lack of complementary viewpoints prevents the students from correcting each other's mistakes, rendering them prone to coupled error accumulation and confirmation bias. In contrast, our proposed heterogeneous architecture combining CLIP and DINOv3 achieves the highest performance of 61.09 mIoU. This superiority confirms that the framework successfully synergizes the global semantic representations of CLIP with the local structural details of DINOv3. By arbitrating between these complementary feature streams, Co2S ensures both semantic correctness and boundary precision, thereby stabilizing the semi-supervised training under scarce data conditions.

3) Effectiveness of Optimization Objectives: To evaluate the individual contribution of each term in our overall training objective, we conduct an ablation study as listed in Table X. The supervised baseline trained solely with supervised loss $\mathcal{L}_{\text{sup}}$ achieves 59.98 mIoU. Incorporating the consistency loss $\mathcal{L}_{\text{ct}}$ boosts the performance to 60.73, indicating that the weak-to-strong constraints effectively mine useful supervisory signals from unlabeled data to augment model training. Furthermore, adding the proposed stability loss $\mathcal{L}_{\text{sta}}$ yields the best performance of 61.09 mIoU. This further improvement confirms that $\mathcal{L}_{\text{sta}}$ is essential for mitigating pseudo-label drift by actively arbitrating between the two students based on confidence, thereby ensuring more reliable mutual learning.

4) Impact of Divergence Formulation on the Stability Loss: The stability loss L_{sta} encourages the two students to produce consistent predictions on high-confidence pixels by aligning their probability distributions. In our implementation, this alignment can be formulated using different divergence measures between the two softmax outputs. To investigate the

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effect of the divergence formulation, we compare several alternatives for L_{sta}, including KL divergence, symmetric KL, Jensen–Shannon (JS) divergence, and their temperature-scaled variants. Temperature scaling ($T = 2$) is used to soften the probability distributions and examine whether smoother gradients improve optimization stability. The comparison results are summarized in Table XI. All divergence formulations achieve similar performance, indicating that the collaborative learning mechanism in Co2S is not sensitive to the specific divergence choice. Temperature scaling slightly improves the KL variant, suggesting that softened probability distributions can provide more stable gradient signals. Among all variants, the mse formulation achieves the best performance (61.04 mIoU) and also exhibits the lowest variance across three runs. These results indicate that the simple mse objective provides stable optimization while maintaining strong segmentation performance. Therefore, we adopt the mse formulation for the stability loss L_{sta} in Co2S.

5) Sensitivity Analysis of the Stability Loss Weight: The stability loss L_{sta} is controlled by a weighting coefficient λ_{sta}, which follows a cosine ramp-up schedule during training. Specifically, λ_{sta} increases from 0 to a predefined maximum value λ_{sta}^{max} within the first 20% of the training iterations and remains constant thereafter. To examine the influence of this parameter, we conduct a sensitivity analysis by varying λ_{sta}^{max}. The results are summarized in Table XII. Co2S demonstrates stable performance across a wide range of values, indicating that the training process is not highly sensitive to the choice of λ_{sta}^{max}. When λ_{sta}^{max} is too small (e.g., 0.1), the mutual learning signal becomes weak, resulting in slightly lower performance. Conversely, excessively large values (e.g., 0.7) introduce overly strong cross-model constraints, which amplify occasional pseudo-label errors and lead to a slight degradation in segmentation accuracy. The best performance is achieved at $\lambda_{sta}^{max} = 0.5$, which is therefore adopted in our final configuration.

6) Sensitivity Analysis of the Confidence Threshold τ: The confidence threshold τ is a key hyperparameter in pseudo-label-based semi-supervised learning, which determines whether a pixel prediction is selected as a pseudo-label according to its confidence score. To investigate its influence, we perform a sensitivity analysis by varying the confidence threshold τ from 0.70 to 0.99. The results are summarized in Table XIII. As τ increases, the pseudo-label accuracy improves because only more confident predictions are retained, while the pseudo-label coverage decreases since fewer pixels satisfy the confidence requirement. Despite this tradeoff, the segmentation performance remains relatively stable, with the mIoU fluctuating within a small margin across different thresholds. Considering the stable performance and for fair comparison with previous semi-supervised segmentation methods such as FixMatch [16] and UniMatch [17], we adopt $\tau = 0.95$ in all experiments.

7) Synergizing CLIP With Different SSM Priors: To assess the advantage of heterogeneous model collaboration, we evaluate CLIP [30] paired with a range of SSMs as well as with itself as a homogeneous baseline. The baseline attains an mIoU of 60.78, representing the performance ceiling achievable by relying solely on vision-language priors. As reported in Table XIV, combining CLIP with any of the examined SSMs [37], [39], [40], [41], [42] yields a performance gain over this baseline. The smallest improvement is 0.06 with MAE [39], while the largest improvement is 0.31 with DINOv3 [37]. This consistent uplift across all pairings confirms that complementary visual priors from diverse self-supervised paradigms substantially enhance mutually semantic interactive learning beyond what CLIP alone can deliver.

TABLE XIV
ABLATION STUDY OF SYNERGIZING CLIP WITH DIFFERENT SSM PRIORS ON THE WHDLD DATASET (1/24)

SSM	mIoU
MAE [39]	60.84
BEiTv2 [40]	60.98
iBOT [41]	60.96
SimMIM [42]	60.99
DINOv3 [37]	**61.09**

TABLE XV
COMPARISON OF PARAMETER COUNTS AND COMPUTATIONAL COMPLEXITY BETWEEN CO2S AND REPRESENTATIVE METHODS

Methods	Trainable Parameters	Inference Parameters	FLOPs
FixMatch [16]	59.5M	59.5M	24.0G
U2PL [19]	66.8M	66.8M	86.4G
UniMatch [17]	59.5M	59.5M	24.0G
DWL [22]	43.9M	43.9M	43.5G
MUCA [29]	27.4M	27.4M	14.2G
Co2S (Ours)	61.2M	88.4M	30.3G

8) Comparison of Parameter Counts and Computational Complexity: To evaluate the model efficiency, we analyze the parameter counts and computational costs of Co2S compared to representative baselines. The floating-point operations (FLOPs) are measured using an input image size of 256×256. As presented in Table XV, Co2S maintains a competitive balance between complexity and performance. Specifically, our method utilizes 61.2 M trainable parameters and operates at 30.3 G FLOPs. While this computational cost is marginally higher than simple consistency-based methods like UniMatch (24.0 G), it remains significantly more efficient than complex pseudo-labeling frameworks such as U2PL (86.4 G) and DWL (43.5 G). Considering the substantial performance improvements, this moderate complexity is well-justified for practical applications.

V. CONCLUSION

We propose Co2S, a stable semi-supervised framework for RS segmentation that leverages complementary priors from two heterogeneous vision foundation models: a VLM (e.g., CLIP) and a SSM (e.g., DINOv3). By co-guiding and co-fusing their inherently distinct representations, Co2S establishes a drift-resistant mutual learning process that

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simultaneously ensures semantic consistency and boundary accuracy. Experiments on six benchmarks demonstrate state-of-the-art performance, with especially pronounced gains under extreme label scarcity.

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